Roll No.: CB.EN.U4AID24[illegible]

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – March 2025

Second Semester

Artificial Intelligence

23AID112 – DATA STRUCTURES AND ALGORITHMS

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Choose an appropriate data structure as applied to a specified problem
CO02	Use various techniques for representation of the data in the real world
CO03	Develop applications using data structures
CO04	Test the logical ability for solving problems

Instructions: Answer all questions. Show all relevant steps and logic according to sections and marks.

PART A [5 * 3 = 15 Marks]

1. Define primitive and non-primitive data structures with examples. [3] [CO01] [BTL 1]
2. Analyse the time complexity of accessing an element and all the elements in a 1D and 2D array? Justify your answer. [3] [CO01] [BTL 4]
3. Given an array arr = [13, 25, 7, 12, 9], Illustrate the first pass of bubble sort and outline the pseudocode for the logical part. [3] [CO01] [BTL 2]
4. Inspect the below pseudo-code. Find the purpose and the time complexity.

```
Algorithm Mystery(arr):
  Input: arr[] - An array of size n
  Output: ????
  sum = 0
  for i = 0 to n-1 do:
    for j = i+1 to n-1 do:
      sum = sum + (arr[i] + arr[j])
  return sum
```

[3] [CO04] [BTL 4]

5. Construct a case-sensitive function to count the vowels in any given sentence [3] [CO02] [BTL 2]

PART B [5 * 5 = 25 Marks]

6. Compare and contrast selection sort and insertion sort in terms of comparisons and swaps. Also, give an input case where insertion sort performs better. [5] [CO02] [BTL 2]
7. Outline the pseudocode for displaying the elements in a Circular Doubly Linked List from any given index both in forward and backward manner. [5] [CO01] [BTL 2]
8. Predict the output of the singly linked list after performing the following operations with pseudocode.

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a. Insert 10 at the beginning
b. Insert 20 at the end
c. Delete the head node. [5] [CO02] [BTL 2]

9. Solve the time complexity of Merge Sort. [5] [CO02] [BTL 3]
10. Examine and justify that the following statements are true:
    a. 17 is $O(1)$.
    b. $n(n-1)/2$ is $O(n^2)$.
    c. $\text{Max}(n^3, 10n^2)$ is $O(n^3)$.
    d. $\sum_{i=1}^{n} i^k$ is $O(n^{k+1})$ and $\Omega(n^{k+1})$ for integer k.
    e. If p(x) is any k-th degree polynomial with a positive leading coefficient, then p(n) is $O(n^k)$ and $\Omega(n^k)$. [5] [CO01] [BTL 5]

PART C

11. Outline the sorting algorithm which works based on partition strategy. Apply the sorting process to the following set of numbers [13, 81, 92, 43, 31, 65, 87, 26, 75] and illustrate properly. [10] [CO02] [BTL 3]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**19**	BTL 1	**3**
CO02	**28**	BTL 2	**21**
CO03	-	BTL 3	**15**
CO04	**3**	BTL 4	**6**
		BTL 5	**5**
		BTL 6	

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